

The Competitive Advantage of Physical IP

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Understanding Consumer Needs

What the consumer
“buys”

Bridging the
divide

What the consumer
“wants”



Performance

Power

Area/Cost

Flexibility

Take photos and video
Play multimedia and games

Without sacrificing quality !

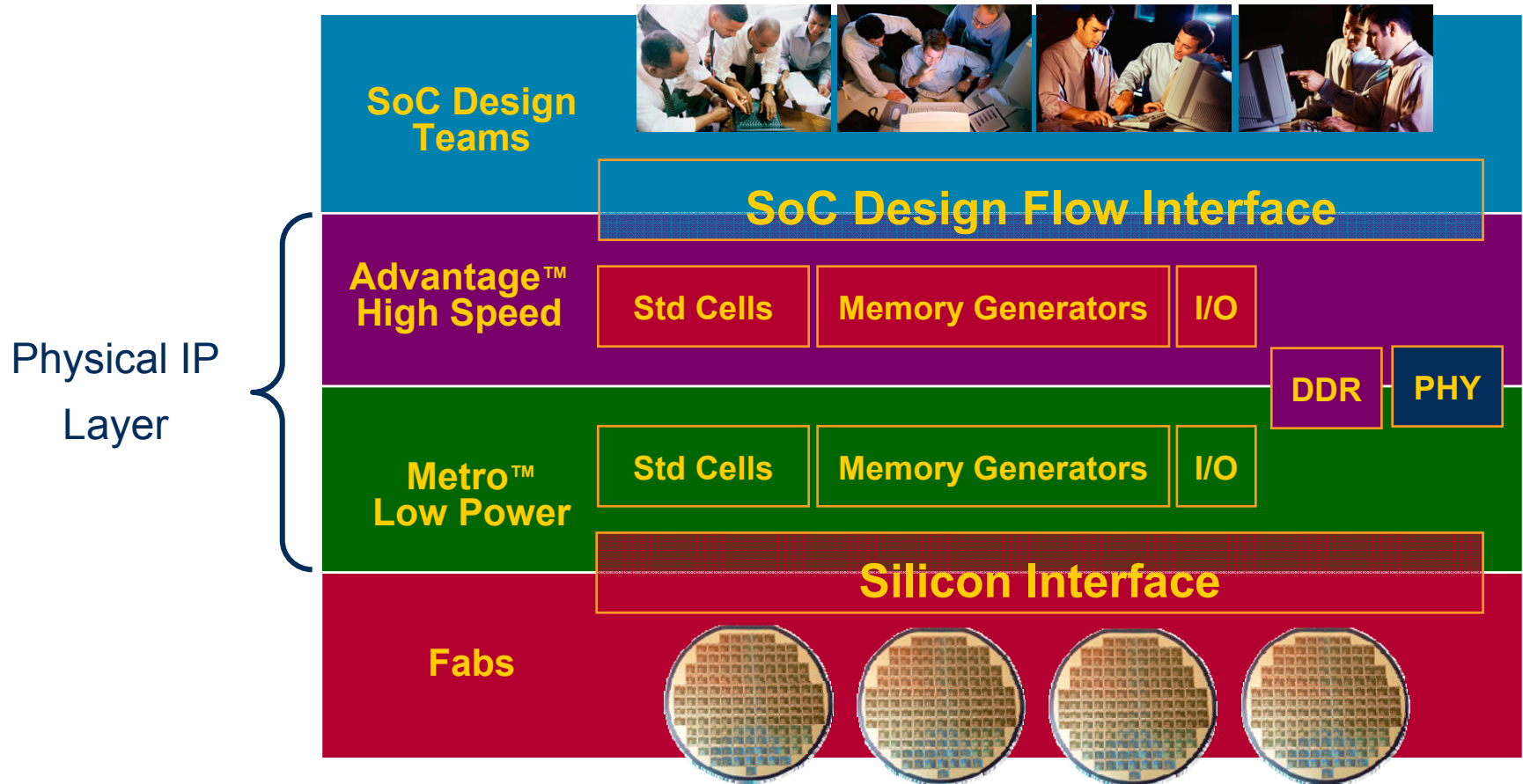
Increased talk time
Play multimedia and games
Send many messages

**Without frequent
recharging !**

All the above ... and
**It should all fit in a pocket
and cost less every time!**

ARM processors + physical IP help bridge the divide

Physical IP Layer

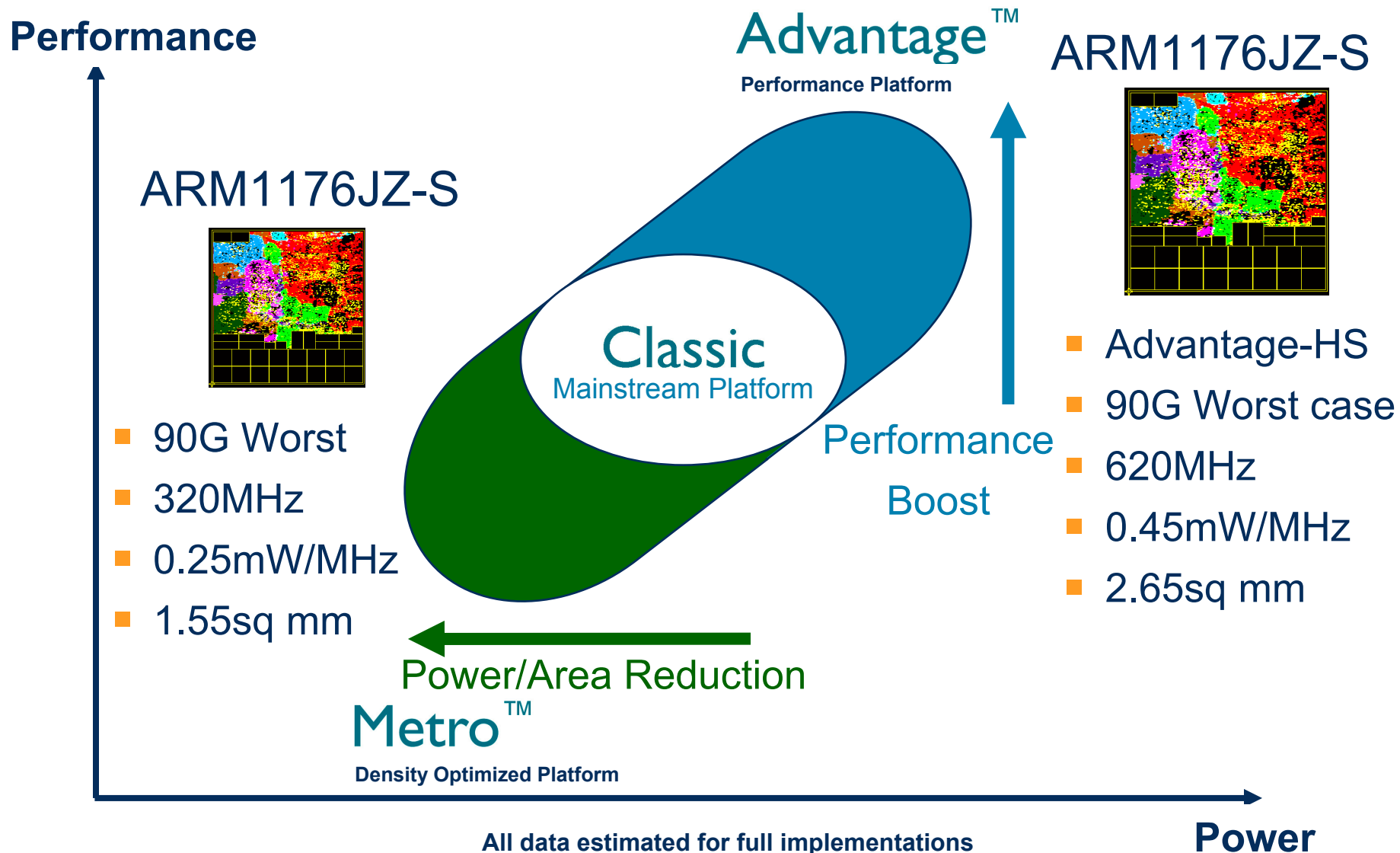


Physical IP Design Complexity

	180nm	130nm	90nm	65nm	45nm
Performance					
Area					
Dynamic Power					
Static Power					
DFM					
Device Variability					

Allows SoC design team to focus on architecture and SoC implementation

Performance/Power Flexibility



Flexibility through Physical IP Platforms

Classic
Mainstream Platform

Advantage™
Performance Platform

Metro™
Density Optimized Platform

Velocity™
High-Speed PHYs

SAGE-X™ Standard Cell	Advantage™ Standard Cell	Metro™ Standard Cell	PCI-Express
Single/Dual Port SRAM	Single/Dual Port SRAM	Single/Dual Port SRAM	Serial-ATA
1/2-Port Register File	1/2-Port Register File	1/2-Port Register File	XAUI
ROM Via or Diffusion	ROM Via or Diffusion	ROM Via or Diffusion	DDR/DDR2/GDDR3
GPIO Inline/Staggered	GPIO Inline/Staggered	GPIO Inline/Staggered	
Power Management Kit	Power Management Kit	Power Management Kit	
Analog/Mixed Signal PLLs, DLLs Specialty I/Os	Analog/Mixed Signal PLLs, DLLs Specialty I/Os	Analog/Mixed Signal PLL, VREG, OSC, Specialty I/Os	

Used in Processor Implementations

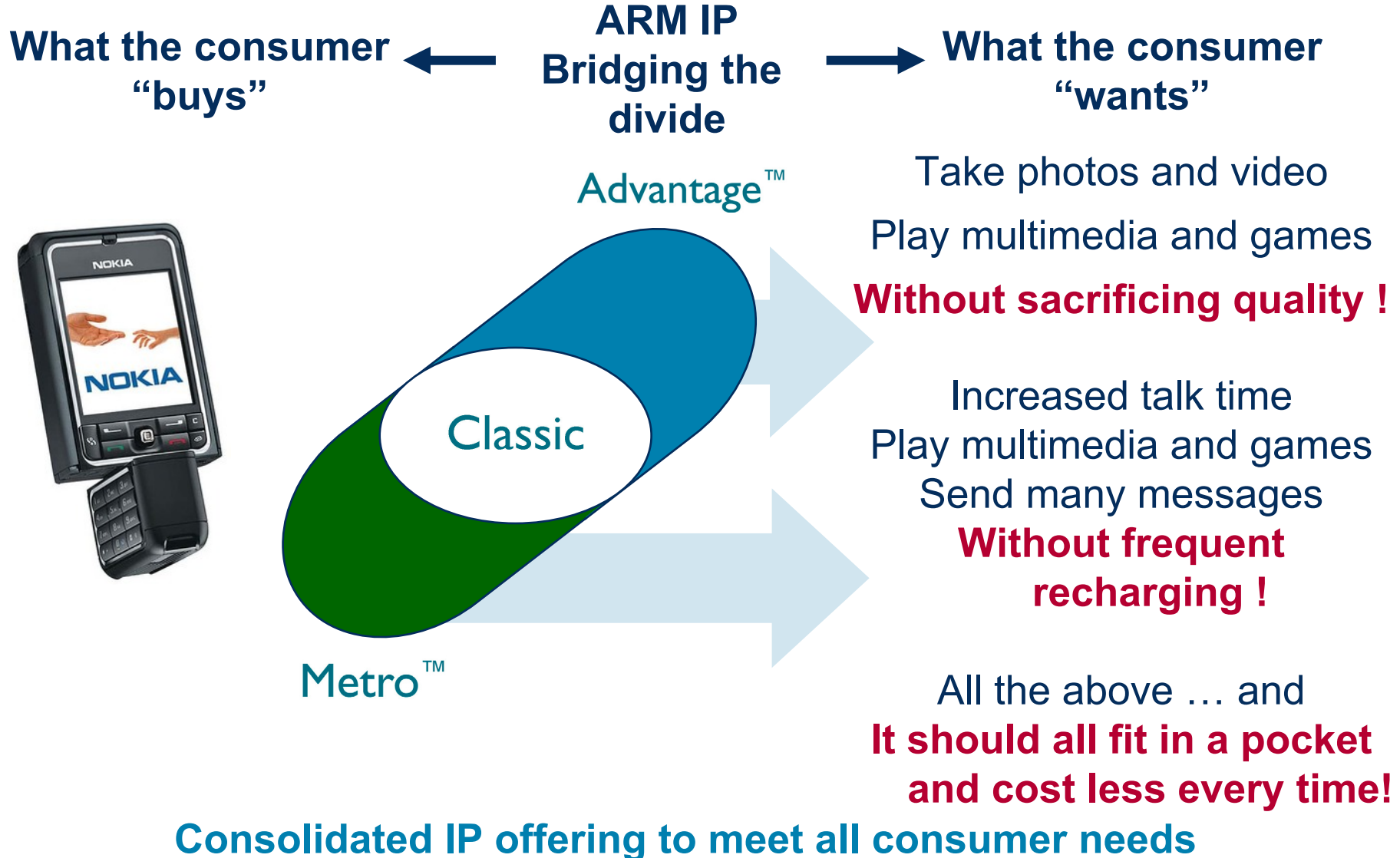
Complete SoC Physical IP Solution

Enabling Power Optimisation

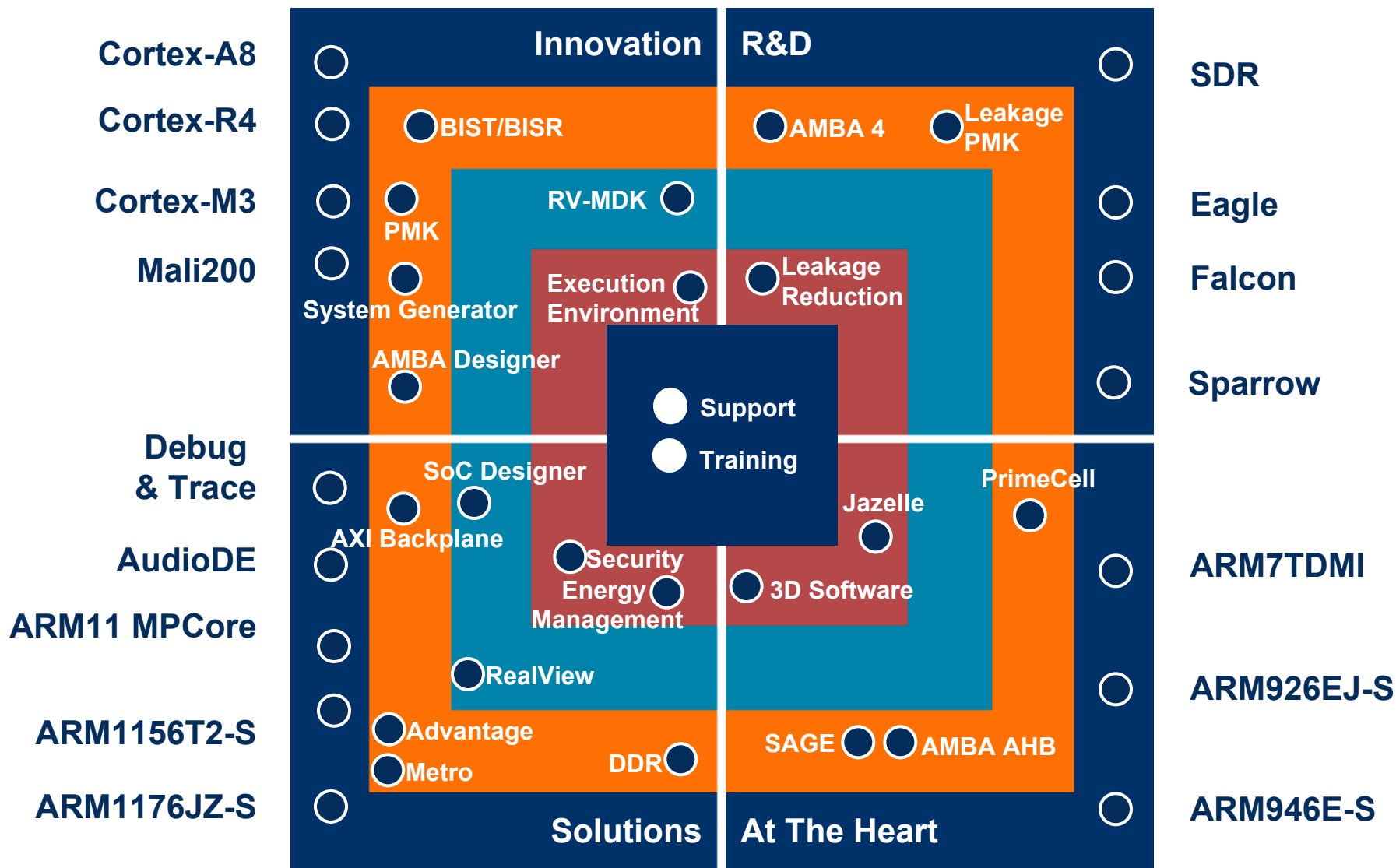
	Metro	Advantage
Dynamic Power control		
Clock gating standard cells	✓	✓
Voltage level shifters	✓	✓
Extended operating range of standard cells and memories → DVFS support	✓	✓
Leakage Power control		
Multi V_T Support in standard cells	✓	✓
Power gating of standard cells	✓	✓
Retention flip flops	✓	✓
Power gating and power down modes in memories	✓	✓*

*65nm products only

Understanding Consumer Needs



Mobile



Summary

- Physical IP Layer
 - Provides EDA views required by SoC design flows
 - Closely linked to process technology
 - Hides process complexity from SoC design teams
- Advanced SoC balancing act
 - Performance
 - Dynamic/static power
 - Area/cost
- Multiple Physical IP Platforms for diverse mobile applications
 - Metro area/power-optimised libraries
 - Advantage performance-optimised libraries
 - CPU-optimised libraries

